

PMMC — Permanent magnet & moving coil
 → D'Arsonval meter, DC & AC

$$m = \frac{I}{I_m}$$

current ranges
 for Range — Resistance
 value

$$(I_2 - I_m) \times R_{sh2} = I_m (R_m + R_y)$$

$$R_{sh1} = R_{sh2} + R_y$$

$$I_{sh} = I_2 - I_m$$

value

$$\begin{aligned} (I_2 - I_m) R_{sh2} &= I_m (R_m + R_{sh1} - R_{sh2}) \\ &= I_m (R_m + R_{sh1}) \\ I_2 R_{sh2} &= \frac{I_m (R_m + R_{sh1})}{I_2} \end{aligned}$$

$$R_{sh2} = R_{sh3} + R_3$$

$$R_{sh3} = R_{sh4} + R_2$$

$$R_{sh4} = R_1$$

$$m = \frac{V}{V_m}$$

$$m = 1 + \frac{R_s}{R_m}$$

$$\frac{V}{V_m} = \frac{R_m + R_s}{R_m}$$

$$V = \frac{V_m}{R_m} (R_m + R_s)$$

$$R_s = \frac{V}{I_m} - R_m$$

~~V = IR~~

$V \propto R \rightarrow dh$

$$S = \frac{R_m + R_s}{V_{range}}$$

$$S = \frac{1}{I_m}$$

VDR $\left(\frac{VR_s}{R_T} \right)$